

- CHAPTER 2 -

PHYSIOLOGY OF STRENGTH AND HYPERTROPHY

WHAT IS STRENGTH?

Let's assume that you are reading this book because you want to know how to train bodyweight strength rather than endurance. At the very least, your goals are to increase your strength to work on the gymnastic isometric exercises such as the planche or front lever, or you want to be able to use your bodyweight strength for various disciplines such as gymnastics, parkour, wrestling, martial arts, MMA, and the like. These goals are important because knowing how your body will respond to the stress will be advantageous when you start to construct your routines. Overall, strength is predicated on a simple equation:

Strength = Neural adaptations * Muscle cross sectional area.

The force output of a muscle is based on the cross-sectional area of the muscle, angle of attack on the joint, individual limb length, and, most importantly, neural factors. Unfortunately, we cannot control the angle of muscle insertions or limb length, which is why they are excluded in the above equation. Developing strength with focus on these neural factors in conjunction with muscle mass gains will provide faster results. In the following sections, we will go through some of the basic physiology and principles that underlie this equation. *Neural adaptations* are what the majority of strength training is about. However, there is some overlap between developing neural adaptations and *muscle cross-sectional area*. This is essentially a fancy term for *hypertrophy* (increases in muscle size). The bigger a muscle is, the stronger it will be.

One of the main questions brought up by those beginning bodyweight training is the intuitive sense that excessive weight will negatively affect their ability. In particular, extra muscle weight seems to be detrimental to bodyweight to strength ratio. This fear—for the most part—is unfounded. Most of the strength athletes—such as gymnasts—and sports with weight classes have athletes with large amounts of muscle mass comparative to their size. Excessive muscle mass tends to have a net negative effect only when you start to become extremely large; bodybuilder size. This usually means, unless you are taking performance-enhancing drugs, you can never have too much muscle mass. Even in weight-class sports you will see that athletes tend to be shorter and have more muscle mass than taller, with more muscle mass. Thus, training toward both strength and hypertrophy is a non-issue in the long run, as both of these attributes will overlap with each other to develop maximal strength and bodyweight to strength ratio.

THE CENTRAL NERVOUS SYSTEM, MOTOR UNITS, AND MUSCLE FIBER TYPES

Motor units are composed of a motor neuron and all of the muscle fibers it innervates. Innervation is the pathway from the motor neurons in the brain to the muscle fibers. The signals are sent to the muscles via electrical impulses. A single motor unit may innervate many different fibers within a muscle, but only innervates muscle fibers of one of the three types.

Motor units are categorized into a continuum that is similar to the muscle fiber types. At one end, you have low threshold motor units (LTMUs); at the other, you have high threshold motor units (HTMUs). LTMUs correspond to the motor units that innervate the Type I slow-twitch fibers and HTMUs correspond to motor units that innervate Type IIX fast twitch fibers. Between those are medium threshold motor units (MTMUs) that innervate Type IIA muscle fibers. The reason why they are termed low versus high threshold is the amount of activation potential (the electrochemical signal in the brain) it takes to make them activate. The easiest way to conceptually understand this is by the strength-endurance continuum from the previous chapter. LTMUs are more *endurance*-geared, and HTMUs are more *strength*-geared.

LTMUs innervate Type I fibers, which are your *slow-twitch* fibers, and are red in color because of the enormous quantity of mitochondria within them. They have a high capacity for endurance and are the primary muscle fiber type that endurance sports develop. These fibers have the least potential for hypertrophy.

MTMUs innervate Type IIA fibers, which are your *adaptable* fibers, are pink in color. They have characteristics of each of the Type I and Type IIX fibers. Therefore, the type of training can bias these fibers toward either strength and power or endurance. This is why training must be specific to your sport. For example, doing high repetition endurance work when your sport is sprinting (which requires a high power output) will adapt your muscles toward endurance. Incorrect training toward the wrong attribute will make you perform more poorly than your competitors. Specificity in sport is king.

HTMUs innervate Type IIX fibers, which are your *fast-twitch* fibers, and are white in color. They fatigue very rapidly because they can only use anaerobic metabolism to supply their energy, but they also contract very rapidly and are the main type of fiber developed in strength and power sports. These fibers have the most potential for hypertrophy.

The fibers are termed *slow-twitch* and *fast-twitch* because of the rate that they produce force when they contract, not solely because of their energy sources. The fast twitch fibers have the greatest potential for hypertrophy, and they also have the greatest potential for strength and power output.

The Henneman's size principle states that motor units are recruited from the smallest to largest. The LTMUs are considered the smallest and the HTMUs are considered the largest because of their physical size. Additionally, due to the size difference, it takes a lower activation potential to activate LTMUs compared to HTMUs. LTMUs are composed of motor units that activate when the required force to move an object is small. Conversely, HTMUs are only activated when the force requirement is high. For example, LTMUs are activated when we want to lift a small object such as a cup, but HTMUs are only activated when we need to use most of our strength, such as in a near-max effort movement such as lifting heavy furniture. Note that during near-max effort or max effort movements the LTMUs are activated along with the HTMUs.

What this means is, when training for strength and hypertrophy, you generally want to use weights that are heavy or bodyweight exercises that are intense and difficult. We want to preferentially increase the rate of growth and development of HTMUs because they have the greatest capacity for the qualities of power, strength, and hypertrophy, as mentioned earlier. Likewise, we want the MTMUs to be mainly trained toward power, strength, and hypertrophy, since they will take on the qualities of the type of training that is imposed on them, which makes them adapt to express HTMU qualities.

This type of training is performed by moving weights or your bodyweight close to maximal intensity, or with lower-intensity exercises performed rapidly with acceleration. This is the importance of the tempo of the exercise. If the weight is heavy or your bodyweight movement is tough and you can only move slowly, you want to focus on performing the repetitions with good form and technique, as fast as possible. Under higher-intensity movements it is your intent to exert maximal force, even if you are moving slowly. This is the reason why I recommend a fast or accelerating tempo in the concentric phase of the movements, in order to ensure that HTMUs are being fatigued and the MTMUs are being trained to be like HTMUs—to induce gains in power, strength, and hypertrophy. If the intensity is light, then you want to perform the exercise with the intent to accelerate your body or weight through the movement, in order to optimally increase strength and power. This will be explored more in-depth in the section about tempo when constructing a workout routine.

There is some new evidence coming out that training with high repetitions may also confer solid strength gains as long as you train with high intensity every other week. This may be useful for those who are having issues with overuse injuries or want to work with high repetitions more than low repetitions.

NEURAL ADAPTATIONS FOR STRENGTH

There are six primary ways that the nervous system increases strength, aside from the hypertrophy of muscles. These will be the primary adaptations in any strength program, so it is important to understand how they work.

- **Recruitment** is an increase in the number of motor units being activated for a specific movement.
- **Rate Coding** or **Firing Rate** is a decrease in the time between each electrochemical signal sent to the corresponding musculature, which increases rate of contraction.
- **Synchronization** or **Intra-muscular Coordination** is a decrease in the amount of time between motor units firing and working together.
- **Contribution** or **Inter-muscular Coordination** is how effectively timed the different contributing muscles to a movement are fired.
- **Antagonist Inhibition** or **Reciprocal Inhibition** is a reduction of resistance from the muscles opposite of those performing the movement.
- **Motor Learning** refers to the neural connections and programs within the brain that will affect your development of learning movements.

We will discuss each of these and their implications for training. You may have heard the term “strength is a skill”—all of these components make up the neural adaptations by which strength is a skill. Some of these components have broad range specificity whereas some do not.

Recruitment increases as force requirements increase. The nervous system has limiters on the amount of force we can produce. Specific structures called Golgi tendon organs in our musculotendinous junctions

(where the muscle starts to become tendon) provide feedback to the brain, which decreases muscle forces to prevent injury in untrained people. Fortunately, with training, the inhibitory effect on force development can be reduced which increases recruitment of muscle fibers. This effect is greatly enhanced and maximized around 85-90% 1 repetition maximum (RM) threshold or approximately 3 RM. Thus, if your primary aim is to gain strength on an advanced program, you will often be programming in exercises in the 1-3 RM range (or close to that repetition range).

Rate Coding or **Firing Rate** increases begin to occur after all motor units in a muscle are recruited due to maximal recruitment or fatigue. When the nervous system senses the recruitment of all available motor units, it further increases strength by sending more rapid electrical signals to the muscles to tell them to contract faster. For most large muscles, such as those used mainly for locomotion, this occurs at approximately 90-92% of 1RM (or 3 RM). In many of the finer motor muscles located in the forearm, rate coding may start to occur as low as 50% of 1RM. Postural muscles—which work constantly in the core and support muscles such as the calves—also rely highly on rate coding for improvements in performance.

Rate Coding matters very little for a strength-based program, but is useful to note for those seeking hypertrophy. Muscles that rely more on rate coding tend to be composed of a greater percentage of slow-twitch fibers. Therefore, they respond better to higher repetitions. Thus, the forearms, calves, core, and other highly-rated coded muscles tend to respond better to higher repetitions when you are seeking hypertrophy. On the other hand, two-joint muscles such as the hamstrings, biceps, and many of the larger muscles, such as the glutes, tend to respond best to more difficult exercises with fewer repetitions because of the preponderance of fast-twitch fibers. Ultimately, if your goal is massive amounts of hypertrophy you may need to alternate your repetition range, rest times, and other factors if one style of training does not seem to be working effectively.

Synchronization or **Intra-muscular Coordination** refers to the nervous system's ability to organize the muscle fiber contractions to make the system more efficient. In untrained individuals, the nervous system recruits motor units in a random or staccato pattern in order to provide the force necessary for a movement. As we further train a movement the motor cortex is able to synchronize the firing of motor units. Imagine a game of tug-of-war. When a team pulls together in synchronization the force is much greater than each person pulling by himself, out of sync with everyone else. This is how the body becomes more efficient when trained. Skills and exercises that are repeated often show the greatest increases in recruitment and synchronization. This is consistent with practicing skills many times throughout the week and repeating exercises multiple times a week over the course of a program.

For example, in the book *Starting Strength*, Mark Rippetoe suggests that beginners perform the squat three times per week. More advanced strength programs, such as Bulgarian Weightlifting Protocol, may have their athletes performing the Olympic lifts as many as two to three times per day, six to seven days a week. Many other sports, including gymnastics, running, swimming, and the like, benefit from massive amounts of technical practice because they require optimal recruitment and synchronization to perform at the highest level. This is true for every sport. Michael Phelps swims miles every day, even though his events are only about 400m in length. Simply put, if we want to get really good at something we have to do it a lot. This will be an important thing to remember when we start to construct routines.

Contribution or **Inter-muscular Coordination** is essentially how effectively you correctly perform a technique. This is the practice part of training that is specific to the movement you are working. For example,

when performing a pull-up you may start the movement with relaxed shoulders. Contribution or inter-muscular coordination is what the body uses to sequence all of the scapular muscles to tighten, which provides a stable base for the shoulders to tighten and then move upward, out of the bottom position. When you are fairly new to exercise your brain may do this quite inefficiently, which is why it is important to learn proper technique in exercises in order to progress safely and effectively.

Antagonist Inhibition or *Reciprocal Inhibition* can improve contraction of the muscle. This is usually performed by extensively stretching the opposing muscle you are planning to work beforehand. The reflexes operate similarly. For example, when the doctor hits your patellar tendon with a reflex hammer, the leg kicks out. This is called *reciprocal inhibition*—where the nervous system activates the quadriceps to fire while simultaneously inhibiting the hamstrings from firing. Thus, you can harness this phenomenon to increase contractions in particular muscles. In particular, paired sets—where you alternate between a pushing movement and a pulling movement—are effective at eliciting this response because of the natural relaxation of the muscle after it has been fatigued.

Motor Learning occurs automatically in the brain and is mostly active for movements that are practiced repeatedly. It occurs all over the cortex in motor planning, the primary motor cortex, cerebellum, and other parts of the brain that are involved in performing movement. This is a primary adaptation of skill work, but it is arguably impossible for one to consciously train it, as the body automatically performs in adaptation to your conscious training. Therefore, we need not go into detail about it. What is important to know, to obtain the benefits of this process, is that you should concentrate 100% while practicing your movements. This will ensure that you are performing them correctly, thus teaching your body the correct movement patterns. If you mess around instead of concentrating with your training, your body will automatically learn sloppy techniques and movement patterns. It has been said “practice makes perfect.” However, it is more accurately stated “*perfect practice makes perfect.*”

THE ROLE OF THE CENTRAL NERVOUS SYSTEM

The central nervous system (CNS) governs the activation of motor units through a variety of systems that are involved with motor planning, activation, and proprioception—the feedback from the body to the brain about where the body parts are in space and the body’s control of them. Since we are not going to look at this in detail, I will say that the CNS, like the muscles, has a set point at which it must be stressed in order to bring about adaptations.

The CNS has a set amount of recovery time that it needs to operate at full capacity. Think of it like a swimming pool. Every time you exercise, you take out some water. Conversely, every time you sleep, take a rest day, eat well, and engage in relaxation or recovery methods not only do you put a little water back in, but you also make it a little deeper and a little further across. Over time, the size of your pool and, therefore, your capacity for water will increase. This is the origin of the term *work capacity*. When you take out too much without replacing it, bad things start to happen. This would be where an athlete starts delving into the overreaching/overtraining realm, which is where progress and performance may plateau or possibly even decline.

Well-structured programs for more advanced athletes have overreaching built into them. Overreaching is planned training beyond a plateau that results in a reduction of abilities such that, after a rest period is taken, the body will adapt and come back with improved abilities over the previous baseline. For example, after a program is completed and a deload week (also called a *recovery week*) is taken, an athlete usually comes back stronger and/or faster. This athlete's pool has increased in capacity during the program, but the water inside of it is not fully replaced until the deload week is taken.

This is worth mentioning because some exercises are more taxing than others. For example, in weightlifting, deadlifts are more taxing than many other exercises because of the large amounts of musculature that are activated during the movement. A *deadlift* is an exercise where there is a bar on the ground, you grip it, and then stand up with it in your hands. It causes significantly more fatigue than most other exercises. This is why deadlifts are placed at the end of many beginner-level programs. Performing them near the beginning would significantly detract from the other exercises in the program because of accumulated fatigue.

CNS Fatigue is an ill-defined concept because there have been no overarching physiological explanations. It could be related to willpower, which is a finite resource. It may even be related to neurotransmitter depletion. For example, we know that taking tests for six hours a day is mentally draining and exhausting. The same is potentially true of extremely intense exercise that activates and fatigues large portions of the brain that is involved in performing movement. We know that high-level powerlifters cannot perform heavy deadlifts multiple times a week. High-level sprinters cannot sprint at full intensity multiple times a week. No high-level athlete can give 100% the majority of the time while training. This is why their routines are constructed to peak during competitions.

As more research continues, it is safe to expect the mechanisms to be further clarified. In the meantime, you should take them with a grain of salt, knowing there is still some uncertainty to what CNS Fatigue actually entails—whether it is neurotransmitters, cytokines, or other factors. Despite the lack of science, good conclusions can still be drawn based on what works in practice.

In bodyweight training, this is analogous to working with supramaximal eccentrics and isometrics. Eccentrics refer to the component of an exercise where the muscles are lengthening, and *isometrics* are exercises where the body is held in one position—without movement—while exerting strength. The *supramaximal component* means that these exercises are too difficult for one to perform compared to a typical repetition. For example, if an athlete is too weak to perform a pull-up for a full repetition they may train utilizing eccentrics by jumping up to the bar and lowering down slowly. Likewise, isometrics may be used in any weak point of the exercise, such as holding the position at the top of the bar (if that is a weak point). All of these tend to fatigue the body more than typical concentric repetitions. Thus, when working with many of these types of exercises each week—or even in a single session—one has to be aware that a plateau in progress may indicate that the best way to achieve progress may be more rest instead of more exercise, which is contrary to what beginners often think when they begin working out.

Later on in this book we will explore constructing programs and how to build in deload periods every four to eight weeks so that you can fully recover from fatigue. These deload periods will double as rest periods, which will allow connective tissues (tendons, ligaments, etc.) some time to heal, as they are typically the first types of tissues to be affected by overuse. When you begin working toward advanced bodyweight movements, two of the most important factors you will need to take into account will be fatigue recovery and connective tissue recovery.

MECHANISMS OF HYPERTROPHY

In the human body, there are three different primary pathways that lead to hypertrophy. The first is mechanical tension. The second is eccentric damage via the popping sarcomere theory and microtrauma. The third is metabolic accumulation, local growth factors, hypoxia, and glycogen depletion based hypertrophy.

Mechanical Tension-based hypertrophy tends to be activated via high intensity exercises like heavy weights and fast movements. This is sometimes referred to as HTMU or fast-twitch fatigue hypertrophy. When there is enough high-intensity mechanical tension on the muscles, the body adds muscle mass in order to compensate. The opposite of this can be seen in the case of a broken bone in a cast: non-movement and non-loading of the nervous system and associated muscle leads to rapid atrophy.

Eccentric Damage and Microtrauma. The intensity of an exercise is heavy enough to create damage to the muscles, but also light enough to perform for enough repetitions to create the damage. The accumulation of repetitions at a certain weight takes a certain amount of time to perform. Many trainers have specified this type of hypertrophy stimulus as *time under tension*—the total amount of time that the muscle needs to be under to adapt by hypertrophy. This is what we discuss as an overall factor for hypertrophy when we talk about “volume” of the total exercises, sets, and repetitions for a particular muscle group.

The damage sets various physiological processes in motion, including satellite cell donation and repair. Satellite cells can be thought of as “muscle stem cells,” which can fuse with damaged muscles in order to help them repair. These are also what contribute to “muscle memory,” where an athlete makes a comeback after taking several years off from training. When they begin training again, they rapidly regain the muscles acquired during their previous period of training. The science behind this shows that the muscles still contain previously fused satellite cells nuclei that aid in rapidly producing the contractile components of the muscle cells again.

Metabolite Accumulation, Local Growth Factors, Hypoxia, and the like. This can be thought of as low-intensity exercise with high amount of repetitions and volume. An example of this type of hypertrophy would be sports that have long, sustained endurance intensity with speed components. For example, cyclists have large quadriceps and rowers have large backs. Another example would be manual labor where one performs a large quantity of light lifting that adds up over the course of a day to create a hypertrophy stimulus. People who work with hammers or plumbers who work with pipes tend to have massive forearms due to low-intensity work they perform consistently over a long period of time.

There are some big misconceptions when it comes to hypertrophy. Specifically, one of the big myths that pervaded the bodybuilding and athletic community for decades is that there was a distinction in the types of hypertrophy that you could build, namely *sarcoplasmic* and *myofibrillar* hypertrophy. *Sarcoplasmic hypertrophy* was thought to occur through utilizing higher repetition ranges such as 8-20 repetitions per set to failure. The specific mechanism of adaptation for this hypertrophy was the metabolic accumulation in the muscle cells. This was exemplified by bodybuilders who tended to have large, but less dense muscles. Conversely, *myofibrillar hypertrophy* was thought to occur through utilizing lower repetitions (such as 1-8) and working more toward strength. The specific mechanism of adaptation for this hypertrophy was the increased accumulation of myofibrillar components of the muscles (such as actin and myosin), which are utilized by the muscles to contract. This was exemplified by strength athletes with muscles that appeared to be very dense, such as olympic weightlifters and gymnasts.

If you are an avid member of the fitness community, you may know that recent research has shown myofibrillar and sarcoplasmic hypertrophy are likely to be a misnomer. There is little if any distinction between them. The main difference would be the strength from those who train with higher repetition ranges and those who train with lower repetition ranges. Likewise, muscle biopsies of the different athletes show that the cellular components of muscle cells will increase proportionally, even in different types of training. Thus, the differences in appearance could be caused by one's hydration level, amount of subcutaneous body fat, intramuscular fat accumulation, or similar factors. For instance, when bodybuilders cut down their body fat for their shows, they certainly have a very dense muscular look, like strength athletes. This is useful information for us because it tells us that the body responds to stress by adding muscle, which can be used effectively regardless of how it was obtained.

One must still consider the importance of the divergent nature of training utilized by strength athletes and those who want to maximize hypertrophy. This involves changing up the programming in regard to frequency of exercise and progressive overload on the muscles. Generally, as you become more experienced, your exercises become more intense. Thus, you may have to increase or decrease frequency based on the particular discipline that you are involved in and the type of practice. In the case of gymnasts, this may be training multiple sessions a day for six or perhaps even seven days per week. However, much of this will be skill work rather than strength work. You may experience positive gains in strength because strength is also a skill. Alternatively, if your goal is maximal hypertrophy, you may want to utilize split routines that put high amounts of volume onto the muscles, thus forcing them to adapt.

Eccentrics and isometrics are of particular importance when discussing bodyweight programming. An example of isometric exercises are the back lever, front lever, and planches. An example of eccentric exercises would be slowly lowering from the top of a pull-up or starting in the top position of a dip and slowly going down. Isometrics are particularly interesting because they branch over multiple pathways toward hypertrophy. In fact, many methods of exercise combine one or two of the different pathways we have discussed. Isometrics are biased toward mechanical tension and metabolic adaptations, while eccentrics are biased toward mechanical tension and eccentric damage hypertrophy the most.

Studies indicate that isometrics and eccentrics tend to recruit HTMUs right off the bat to sustain their contractions. This makes sense, as they are very difficult movements. The irony of eccentrics is that the studies on eccentric training seem to indicate that one-second eccentric movements are better at stimulating hypertrophy and strength adaptations than longer eccentrics, such as six seconds. This means that fast eccentrics preferentially activate the HTMUs, which have the greatest potential for power, strength, and hypertrophy. This understanding is accentuated by one's understanding of physiology. In longer eccentrics, the occluded blood flow to the muscle and greater time under tension also means they are biasing toward metabolic slow twitch adaptations. Likewise, isometric holds that are too short will not give one enough volume to force adaptations. However, isometrics held too long may bias one's adaptations toward higher levels of endurance. What this means for training is that one needs to perform their isometrics in the "sweet spot" ranges in order to maximize strength and hypertrophy benefits. This will be explored at length in later chapters on routine construction.

As research continues these mechanisms without a doubt will be further clarified; so one should take them with a grain of salt. There is still some uncertainty surrounding the factors that lead to muscular hypertrophy.

Here is one of the primary concepts for you to take away from this section: While it is important to know that certain adaptations and differences in hypertrophy mechanisms may occur, there is really no such thing as unwanted hypertrophy—unless it makes one too heavy for their sport or weight class.

The second primary concept you should take away from this section is the fact that overall volume in the context of frequency means the most in regard to hypertrophy. The volume of the exercise on particular muscles must exceed a certain threshold for hypertrophy, which in effect will increase as your muscles get bigger. This is easily seen in the difference between beginners and bodybuilders. Beginners can have significant hypertrophy with relatively few exercises while bodybuilders have tons of volume in their routines. Likewise, frequency plays a role as there are many occupations—such as in construction and furniture movers—where one's work is their only “workout” for the day, but their body will add hypertrophy in response to the muscle stress that comes from repeatedly moving heavy objects.

The concept of ensuring enough volume for hypertrophy in your exercise routine will also be explored as you learn to construct your own custom routine. We will outline some simple concepts on how to work with certain repetition ranges and sets to obtain the right amount of volume. This is why there is no discussion specifically on the “best” repetition ranges in this section, though generally 5-15 repetitions tend to give beginners an adequate amount of volume to achieve hypertrophy.

If you are interested in studying this topic further, Brad Schoenfeld and Mike Zourdos have published some excellent research on hypertrophy and overall volume.

OPEN AND CLOSED CHAIN EXERCISES

Open Kinetic Chain (OKC) exercises are performed in a manner that leaves one's limbs free to move. These exercises involve movements of the limbs in space, weighted or unweighted. The weighted versions of these are isolation exercises (such as leg extensions and hamstring curls on machines). Some examples of OKC upper-body movements are biceps curls and triceps extensions.

Closed Kinetic Chain (CKC) exercises are performed in a manner where one's limbs are not free to move. Typically, these movements are performed with barbells or bodyweight. Weighted versions of these exercises include squats, deadlifts, and the Olympic lifts, where one's feet are fixed against the ground and the body is moved against it. Likewise, almost all bodyweight movements are closed chain—the hands or feet are fixed against the ground or other implements. Single leg squats, dips, pull-ups, pushups, and handstand pushups, are all performed with one's hands fixed against the ground, bars, or rings.

Bench and press are barbell exercises that are stabilized in space; this represents somewhat of a hybrid between OKC and CKC exercises, as you stabilize the weight (like many OKC exercises) but your hands are fixed while doing so (like CKC exercises). When you take a look at the strength difference between semi-OKC exercises—such as the press/military press—and a pure CKC exercise—such as the handstand pushup—you will see that the CKC exercise tends to be stronger than the semi-OKC exercise. For example, if you subtract the arm weight from the handstand pushups, you will likely find that you can perform more handstand pushups comparable to the weight you can press. This is likely due to internal factors that result from co-contraction and increases in kinesthetic feedback to the body.

The main thing to keep in mind is that CKC exercises are more applicable for building strength in the upper body. However, it is more difficult to track your progress as it is not easy to reliably measure incremental improvements like adding weights to a barbell. If your goal is hypertrophy, CKC and semi-CKC exercises typically work best. These could include compound barbell exercises (squat, deadlift, bench press, etc.). One interesting phenomena is that the overhead barbell press tends to be a better muscle mass builder than the handstand pushup, which requires significantly more body stabilization, thus limiting the amount of force that can be used in performing the movement. This slightly decreases the amount of hypertrophy gained from the movement due to less mechanical tension.

If your ultimate goal is purely hypertrophy, it is generally a good idea to perform primarily barbell-type exercises. This is not to say you cannot gain an impressive physique with bodyweight exercises, it will just take longer.

For rehabilitation, you will primarily use OKC exercises because they allow you to easily isolate specific weaknesses, target specific structures that need to be strengthened, or improve certain movement patterns. The goal of therapy is to work your way back to gross motor CKC movements. For example, if you are undergoing therapy for an ankle sprain, you will begin by performing mostly OKC exercises, which will help strengthen the muscles in the leg and prevent them from atrophying. However, as you improve range of motion, strength, and other factors you will begin performing more gross movement patterns—such as squats and standing drills—to strengthen the ankle so it can perform athletic movements again. Likewise, with something like elbow tendonitis, you would begin with isolation exercises to rehabilitate the injured area and eventually progress to function-based movement such as pull-ups or other compound exercises. We will discuss this more in the injury sections of this book.

To summarize, closed chain exercises emphasize stabilization in the core and extremities because body position plays a role in its interaction with the ground, parallettes, or rings. This means that bodyweight exercises tend to rely more on progressions rather than adding weight. They work extremely well in the development of strength, proprioception, and kinesthetic awareness. When you can add weight to them—such as with a weighted vest—it makes progressions much easier to handle. The benefit of performing more open kinetic chain exercises is that they can be regulated easier with weights. This is especially true if you have an injury or weakness that may benefit from exercise that is focused on certain muscles, tendons, or other structures.

All these movements are useful in their unique contexts. If you are reading this book you probably have an interest in bodyweight strength, including the various isometric hold positions that gymnastics is known for. Additionally, most people have some aesthetic goals, such as looking good naked. If these are your goals, this book is a good fit for you.

CHAPTER 2 SUMMARY

PHYSIOLOGY OF STRENGTH AND HYPERTROPHY

Knowledge Base

- Strength = Neural adaptations * Muscle cross sectional area.
- Neural adaptations are enhanced when working with the intensity of exercises. These include increased neurological efficiency in motor recruitment, rate coding, synchronization, contribution, reciprocal inhibition, and growth and pruning.
- Muscle cross sectional area is the increase in muscle hypertrophy (muscle growth).
- If your goal is strength *and* hypertrophy, you will want to execute movements as explosively as possible with good technique. This will help recruit and stimulate the HTMUs, which are composed of fast-twitch fibers that carry the greatest potential for strength and hypertrophy.
- Practice does not make perfect; *perfect* practice makes perfect.
- Work capacity is naturally increased as you train. You can think of it as building a larger sized pool to replace a smaller pool. Thus, training and recovery are equally important as training takes away water from the pool and recovery expands the pool and puts water back into it. Training programs should include planned deloads to allow for proper recovery.
- Eccentrics and isometrics can be used extremely effectively in a training program, but they should not make up the majority of your program because they are extremely taxing on the central nervous system.
- Hypertrophy tends to occur through three mechanisms: mechanical tension, eccentric damage and microtrauma, and local metabolic and hypoxic factors.
- There is no such thing as myofibrillar or sarcoplasmic hypertrophy, just hypertrophy.
- Beginners tend to do best with a 5-15 repetition range. For maximal hypertrophy, you will want to work in the 5-20 range and even outside of it, with all sorts of different modifications to your program.
- CKC movements are typically better for generating strength and muscle mass. Specifically, CKC bodyweight movements are better at generating strength, while CKC barbell exercises are better at generating hypertrophy. They do overlap some.
- OKC exercises are good for targeting specific weaknesses or rehabilitating injuries.

Application

It is important to know how the physiological concepts play a role in constructing a workout routine. These will be helpful once you start implementing your routine, because you will be able to use your knowledge of body physiology to problem-solve your routine on your own. For example, if your goal is maximum hypertrophy, you will want to alter the tempo, rest times, exercises, and other factors in your routine to ensure that you continually progress. In the following quote, consider “knowing yourself” to be your body’s physiology and your workout routine to be your enemy.

If you know the enemy and know yourself, you need not fear the result of a hundred battles.

If you know yourself but not the enemy, for every victory gained you will also suffer a defeat.

If you know neither the enemy nor yourself, you will succumb in every battle.

~ Sun Tzu